Akshay Meshiya

Manufacturing Engineer
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PROFESSIONAL SUMMARY

Manufacturing engineer with an M.S. in Chemical Engineering from Rowan University and 6+ years supporting manufacturing operations, new product introduction (NPI), and continuous improvement in chemical and regulated plants. I have developed manufacturing processes, handled scale-up and technology transfer, and solved equipment and process problems on the floor. I rely on statistical process control (SPC), root cause analysis (RCA), CAPA, and clear SOPs to cut defects, cost, and cycle time, applying Lean manufacturing principles and working cross-functional with production, quality, maintenance, and EHS.

WORK EXPERIENCE

Process Engineer **Apr 2026 – Present**
Valcor Engineering Corporation **NJ, USA**

- Develop PFDs, P&IDs, and 3D plant layouts in AutoCAD Plant 3D and SmartPlant P&ID for 12 equipment upgrades that raised unit production capacity by 10%, managed through management of change (MOC).
- Reduce process variability in batch and continuous operations with statistical process control (SPC) and downtime analysis, raising throughput while holding cleanroom-grade purity.
- Build and maintain OSIsoft PI System and Power BI dashboards that track critical KPIs and process trends in real time, speeding up fault detection and cutting unplanned downtime by 15%.
- Support $2.5M+ in CAPEX projects by commissioning ultra-high-purity (UHP) gas and liquid delivery systems and linking them to SAP PP/PM for maintenance visibility and equipment traceability.
- Model high-purity chemical processes in Aspen Plus, Aspen HYSYS, and ChemCAD, improving solvent recovery efficiency by 7% while meeting environmental compliance standards.
- Lead HAZOP and LOPA studies with root cause investigations, identifying 10+ safety and operational risks and keeping the site audit-ready under strict regulatory requirements.

Assistant Production Engineer **Feb 2022 – Jun 2024**
R.R. Chemical (ISO 9001:2015 Certified) **Gujarat, India**

- Improved batch manufacturing processes by analyzing each step and redesigning where needed.
- Supported new product introduction (NPI), scale-up, and technology transfer from R&D and pilot scale to full production.
- Dug into batch interruptions and deviations with root cause analysis (RCA, 5 Whys) and put corrective and preventive actions (CAPA) in place, reducing batch failures by 15%.
- Fixed process and equipment problems on reactors, pumps, and process lines, cutting downtime.
- Ran in-process quality control (QC) checks on yield and specifications under ISO 9001.
- Wrote and updated SOPs, work instructions, MSDS/SDS, and EHS documents to standardize manufacturing work.

Process Executive & Chemist **Jul 2018 – Feb 2022**
Palsana Enviro Protection Ltd. (PEPL) — Common Effluent Treatment Plant **Gujarat, India**

- Ran shift operations at a common effluent treatment plant (CETP) handling wastewater from textile units, across primary, secondary, and tertiary treatment.
- Replaced handwritten logbooks with Microsoft Excel trackers and trend charts for lab results, chemical use, and flow, which cut shift reporting and MIS preparation time by 30%.
- Used jar tests to fine-tune coagulant, flocculant, lime, and polyelectrolyte doses, cutting chemical consumption by 8% and lowering treatment cost per unit of effluent.
- Raised plant capacity utilization by 10% by keeping treatment units within design limits, so the CETP could take effluent from more member industries.
- Watched data trends to catch process deviations early, which helped operations avoid downtime and off-spec discharge.
- Tested inlet, in-process, and treated effluent for pH, COD, BOD, TSS, TDS, color, oil & grease, MLSS, and SVI.
- Checked treated effluent against GPCB/CPCB discharge norms and helped prepare compliance reports and records for regulatory inspections.
- Worked with operations and maintenance on pumps, clarifiers, aeration, and dosing equipment, following chemical handling and EHS procedures.

SKILLS

- **Manufacturing Engineering:** Manufacturing Process Development, New Product Introduction (NPI), Scale-Up, Technology Transfer, Continuous Improvement, Lean Manufacturing Principles, Standard Work, Yield Improvement, Cycle Time Reduction, Cost Reduction
- **Quality & Reliability:** Statistical Process Control (SPC), Root Cause Analysis (RCA), CAPA, ISO 9001, Quality Control, Equipment Reliability, Preventive Maintenance, Downtime Analysis, KPI Monitoring
- **Process & Design:** PFD/P&ID Development, Material & Energy Balances, Process Optimization, Equipment Sizing, Commissioning, Process Control (PID Loops, DCS/PLC Fundamentals)
- **Software:** AutoCAD, AutoCAD Plant 3D, SmartPlant P&ID, Aspen Plus, Aspen HYSYS, ChemCAD, SAP (PP/PM), OSIsoft PI System, Power BI, Minitab, Python, Microsoft Excel (Advanced), Microsoft Office
- **Safety & Compliance:** HAZOP, LOPA, OSHA Process Safety Management (PSM), Risk Assessment, Management of Change (MOC), SOPs, MSDS/SDS, EHS, Regulatory Compliance, Data Analysis

PROJECTS

Manufacturing of Ethylamine — B.E. Final Year Design Project **2021**

- Designed a conceptual 100 TPD ethylamine production facility; performed material and energy balances, equipment sizing, reactor and distillation design, cost estimation, and process safety evaluation.

Manufacturing of Ethylene Glycol — Diploma Final Year Design Project **2018**

- Executed process design, feasibility analysis, material balance calculations, utility assessment, and process economics evaluation.

EDUCATION

Master of Science, Chemical Engineering — Rowan University, Glassboro, NJ **Aug 2024 – May 2026**

Relevant Coursework: Process Optimization, Advanced Reactor Design, Transport Phenomena, Advanced Engineering Thermodynamics, Polymer Engineering, Water & Environment Monitoring, Sustainability Assessment, Strategic Engineering Management

Bachelor of Engineering, Chemical Engineering — Gujarat Technological University **Jul 2018 – Jul 2021**

Relevant Coursework: Process Equipment Design I & II, Mass Transfer Operations I & II, Chemical Reaction Engineering I & II, Process Heat Transfer, Fluid Flow Operations, Chemical Engineering Thermodynamics I & II, Instrumentation & Process Control, Process Modeling, Simulation & Optimization, Multicomponent Distillation, Plant Design & Project Engineering, Pollution Control, Safety & Health Management

Diploma, Chemical Engineering — Gujarat Technological University **Jul 2015 – Jun 2018**

Relevant Coursework: Mechanical Operations, Process Heat Transfer, Mass Transfer, Industrial Stoichiometry, Chemical Process Technology I & II, Utilities & Instrumentation in Chemical Plant, Pollution Control & Effluent Treatment, Safety & Hazard Management in Chemical Industry, Petroleum Refining & Petrochemical Technology